

An Enhanced Deep Neural Network Model for the Detection of Anomalous Behavior of Drivers in Road Traffic

Abstract

Over recent years, video-surveillance systems have seen extensive adoption, largely driven by security imperatives, with radar-based speed detection being a common feature in traffic monitoring. Despite its prevalence, broader anomaly detection in traffic patterns has not received equivalent focus. This research develops a sophisticated deep learning framework, drawing architectural inspiration from MobileNet, ResNet50, and VGG19, to not only detect and track vehicles but also analyze trajectory data to identify nonstandard behaviors. Specifically, our model detects four distinct anomalies: overspeeding, lingering in no-stopping zones, insufficient spacing between vehicles, and violations of traffic light signals. To support this, we constructed a unique dataset comprising over 60,000 video frames. The YOLOv3 algorithm facilitated initial object recognition, which was complemented by data augmentation techniques to mitigate issues related to class imbalance and the limited availability of annotated datasets in this domain. Our enhanced model achieved an overall accuracy of 95%, with a detailed performance breakdown for each detected anomaly.

1. Introduction

The escalation in the availability of video feeds from both public and private sources has significantly boosted research into video analysis and anomaly detection [1]. This surge in technological advancements offers a compelling incentive for developers to create systems capable of identifying and addressing abnormal behaviors in video content swiftly and more effectively. In recent years, artificial intelligence (AI) has ushered in a transformative era in video surveillance. Leveraging machine learning and deep learning techniques, modern video surveillance systems now possess the capability to detect and analyze behaviors with precision and speed unattainable by human observers [2].

Anomalies in video surveillance are defined as patterns in data that deviate from what is typically recognized as normal. The applications of anomaly detection are broad and varied, encompassing areas such as credit card fraud prevention, insurance fraud detection, and network and cybersecurity intrusion monitoring. Additionally, it plays a critical role in diagnosing issues within safety-critical systems and supports military operations by monitoring hostile activities.

The behavior of drivers has long been scrutinized, especially as it relates to road safety [3]. Abnormal behavior by drivers, such as speeding in unsafe conditions, erratic lane changes, illegal maneuvers, and failing to adhere to traffic signals, significantly contributes to road accidents [4, 5]. These incidents not only cause physical and mental distress to those involved but also lead to substantial financial losses due to vehicle repairs and insurance claims. Among the specific behaviors studied, driver

drowsiness has been extensively analyzed through facial expression monitoring, and yawning detection has been explored by classifying the states of mouth openness [3, 6]. Nevertheless, other atypical behaviors like prolonged stops in yellow box junctions and sudden lane changes remain underexplored.

The proliferation of computer vision and AI technologies has catalyzed the development of increasingly sophisticated applications in these fields [2, 7]. Significant progress has been made in the automatic recognition of human movements and gestures, as highlighted in various studies [8–10]. Despite the vast amount of research dedicated to AI, the application of deep learning techniques in the detection of abnormal behavior is still nascent. This early stage of development underscores the ongoing need for exploration and advancement in this area.

The relentless pursuit of developing efficient and effective models for the recognition and classification of abnormal behaviors continues to be a driving force in the field. While substantial progress has been made, the challenge of creating models that deliver robust performance in real-world scenarios remains. This ongoing endeavor not only highlights the dynamic nature of the field but also emphasizes the potential for significant breakthroughs that could further enhance the capabilities of video surveillance systems. The commitment to advancing these technologies promises to yield tools that are not only more capable but also more accessible to a broader range of applications in public safety, security, and beyond.

2. Related Work

The field of anomaly detection has seen substantial growth, driven by advancements in model-based, proximity-based, and classification-based approaches. The ongoing development of sophisticated algorithms and the integration of comprehensive data mining techniques suggest that future systems will be increasingly effective at identifying and responding to anomalies in a variety of domains, enhancing both security and operational efficiency.

2.1. Model-Based Anomaly Detection

Model-based anomaly detection revolves around constructing a mathematical model that characterizes normal behavior, thereby identifying deviations as anomalies. As highlighted by Sodemann, Ross, and Borghetti [1], this approach is pivotal in automated surveillance systems, capturing the nuances of typical activity through statistical models or machine learning algorithms. For instance, Li, Liu, and Huang [11] effectively utilized Gaussian mixture models (GMM) to segregate normal traffic patterns from anomalous ones, proving the model's efficacy in capturing complex distributions in traffic flows. The extraction of local invariant descriptors in this study provides a robust basis for feature selection, which is crucial in enhancing detection accuracy.

Moreover, Colque et al. [12] advanced the model-based approach by incorporating histograms of optical flow orientation, magnitude, and entropy (HOFME) in their analysis. This technique underscores the importance of detailed feature vectors that include orientation and magnitude, offering a refined perspective on motion-based anomalies, especially in crowded and dynamic environments [12].

2.2. Proximity-Based Anomaly Detection

Proximity-based methods identify anomalies by examining the data points' relationship to their neighbors. This method assumes that normal data points occur in dense neighborhoods. Anomalous instances are detected based on their relative isolation from this dense cluster, as detailed by Colque et al. [12]. Li, Liu, and Huang [11] further elaborated on this by introducing outlier scoring based on relative density, providing a quantifiable metric to assess the degree of anomaly, which enhances the interpretability of detection results and facilitates threshold setting for alerts in real-time monitoring systems.

2.3. Classification-Based Anomaly Detection

Classification-based models are extensively used due to their straightforward conceptualization of normal versus abnormal classes within the feature space. Chen, Yu, and Li [13] highlighted the use of multiclass classification to detect varied types of traffic incidents, which enables more specific responses to different anomaly types. This approach is contrasted by the one-class classification method, where all training data are assumed to represent “normal” behavior—a method used by Patil and Biswas [14] to identify visual surveillance anomalies with a support vector machine (SVM).

Additionally, the application of neural networks and autoencoder ensembles by Chen, Yu, and Li [13] demonstrates significant advancements in handling high-dimensional data and complex anomaly patterns, particularly effective in overcoming the limitations posed by traditional methods when dealing with large-scale data.

2.4. Integration of Advanced Data Mining Techniques

The integration of advanced data mining techniques, as explored by Silva et al. [15], represents a pivotal evolution in anomaly detection. Adopting the CRISP-DM framework, these methods facilitate a structured exploration of data, enabling the identification of intricate patterns that may indicate anomalous behavior. Such techniques have shown promising results, particularly in domains requiring high accuracy and rapid processing, such as real-time driver behavior monitoring and automated traffic management systems [15].

2.5. Current Trends and Future Directions

Recent studies, such as those by Matousek et al. [16] and Wang et al. [17], indicate a shift toward leveraging ensemble learning and deep learning techniques to improve

anomaly detection's robustness and accuracy. These methods not only enhance the detection of complex patterns but also offer adaptability in various operational contexts, suggesting a rich avenue for future research [16, 17].

Researchers have adopted various methodologies to detect and classify atypical behaviors, with a particular emphasis on driving behaviors and drivers' conditions. An examination of the extensive literature reveals a significant focus on these aspects. Table 1 provides a summary of the related work. Despite this, there exists a notable gap in the development of comprehensive applications aimed at monitoring drivers' unusual behaviors, such as lingering in no-stopping zones, maintaining improper distances between vehicles, and failing to adhere to traffic signals. Given the increasing demands for enhanced security and efficient traffic management, there is considerable potential for the development of sophisticated automated systems designed to detect and address these kinds of driver anomalies.

Table 1. Summary of existing work.

Authors	Technique	Discussion
Li, Liu, and Huang [11]	Gaussian mixture model	In this work, the local invariant features were extracted. However, only one specific abnormal behavior was extracted
Colque et al. [12]	Histograms of optical flow orientation, magnitude, and entropy (HOFME)	Magnitude and direction were used to create the feature vector after feature extraction. This technique cannot be used for all types of anomalous behavior
Ali, Atia, and Mostafa [18]	<i>k</i> -Nearest neighbor and dynamic time warping algorithms	Only two class classification was carried out. The abnormal behavior of drivers was not categorized into specific class
Silva et al. [15]	Have used random forest (RF) and gradient boosting (GB)	This work extracted the anomalies on the roads, like bumps and holes, among others, with respect to the vehicle

Authors	Technique	Discussion
Matousek et al. [16]	Neural network	Authors detected aggressive, anxious, nervous, and unstable driving with respect to the driving speed only
Wang et al. [17]	Ensemble learning	Have focused only on the vehicle motion parameters. The road environment was not investigated

2.6. Contribution

The development of applications for monitoring driver behavior presents numerous challenges, including vehicle tracking, calculating the duration vehicles remain in unauthorized zones, and assessing driver responses to changes in traffic signals, among others. This research aims to identify and analyze aberrant driving behaviors with respect to road rules compliance and vehicle proximity, such as maintaining distance. Furthermore, convolutional neural networks (CNNs) have shown promising results in feature extraction and classification across various domains. Despite their potential, the application of CNNs in the context of abnormal driving behavior detection remains underexplored, prompting increased interest in their potential uses and effectiveness.

The primary contributions of this study are outlined as follows:

-
-

Detection and extraction of multiple elements from video footage, including various vehicles, traffic lights, and no-stopping zones (yellow boxes).

-
-

Application of transfer learning to our proprietary datasets using architectures such as MobileNet, ResNet50, and VGG19.

-
-

Development of an advanced CNN model tailored for multiclass classification of atypical driver behaviors (such as speeding, dwelling in no-stopping zones,

maintaining inappropriate vehicle spacing, and disregarding traffic signals), surpassing the capabilities of existing systems.

-
-

Comprehensive demonstration and analysis of the impacts of fine-tuning and optimizing hyperparameters to enhance accuracy beyond that of current applications.

3. Materials and Methods

This section details the components developed for the prediction of the anomalous behavior of drivers.

3.1. Architectural Design of the Anomaly Detection Application

The model initiates by extracting frames from video feeds to construct an image dataset representing four distinct classes of driving anomalies: speed, vehicle spacing, duration spent in no-stopping zones, and nonadherence to traffic signals. Initially, relevant objects are detected within these images. Subsequently, vehicle tracking is implemented to monitor ongoing behavior. Features are extracted by a trained CNN, leveraging the architectures of established pretrained models such as MobileNet and ResNet50.

The devised system processes the extracted data to predict various parameters, including vehicle speed, the time elapsed in no-stopping zones, and the distances maintained between vehicles. It also recognizes traffic lights and evaluates compliance with traffic signals by determining whether vehicles encroach upon restricted pathways during a red light. Figure 1 delineates the comprehensive architecture of the proposed system.

Figure 1

[Open in figure viewerPowerPoint](#)

Architecture of the proposed system.

3.2. Dataset Creation

To address the absence of a comprehensive dataset encompassing all anomalies pertinent to this research, we constructed a proprietary dataset tailored to our specific needs. This dataset was developed by extracting data from 60,000 video frames captured under controlled conditions, designed to analyze speed-related anomalies. To enhance the robustness and generalizability of our findings, we varied the lighting conditions across a spectrum from normal to moderate levels.

Moreover, we incorporated a range of meteorological conditions into our dataset, including sunny, cloudy, and rainy scenarios, to evaluate the system's performance under diverse environmental influences. From this extensive video data, we successfully extracted and quantified the speeds of 858 vehicles, comprising a variety of types such as cars, vans, lorries, and buses, which were subsequently cataloged within our dataset.

Additionally, for the analysis of behavior in no-parking zones, we processed 46,089 frames sourced from our surveillance cameras specifically focused on these areas. Within these frames, we detected and tracked 325 vehicles maneuvering within no-parking zones. The system meticulously recorded the entry and exit times for each vehicle, capturing their precise duration within these restricted zones. This comprehensive data collection and processing effort supports the development of advanced analytics for identifying and understanding vehicle behavior concerning designated no-parking areas [19, 20]. Concurrently, the process of extracting distances between moving vehicles was conducted across various designated areas. To ensure accuracy in distance measurement, the system required a one-time calibration in each area due to the variability in scale across different video feeds; for instance, a span of 10 pixels may not consistently represent the same physical distance in every video context.

For the purpose of distance measurement, the system analyzed ~432,000 video frames, successfully identifying over 2500 vehicles. Initially, the extracted distance data were recorded in a CSV file. This preliminary dataset was subsequently processed using a dedicated analytical application, which classified the data into categories of "correct" and "anomalous" based on predefined criteria. The characteristics and specifics of this dataset, including the methodology for classification and the results obtained, are summarized in Table 2, providing a detailed overview of the dataset composition and the analytical outcomes.

Table 2. Details of the dataset.

Anomaly	Number of images/data
Standing on yellow box	1520
Traffic light signs	1810
Distance between vehicles	1070
Speeding	1680

3.3. Vehicle Detection Using YOLOv3 and MobileNet Single Shot Detector (SSD)

The vehicle detection process, a fundamental component in developing this application, has been implemented using advanced deep learning techniques, diverging from the traditional background subtraction methods. Among the prevalent deep learning-based object detection algorithms are faster region CNNs (Faster R-CNNs) [21], you only look once (YOLO) [22], and SSDs [23]. Following a thorough evaluation, the YOLOv3 algorithm was selected for its superior performance in vehicle detection.

YOLOv3 is particularly well-suited for real-time object detection in complex scenes with diverse objects, making it highly effective in environments where vehicles are closely spaced or located in densely populated areas. This algorithm is capable of detecting overlapping small objects, a critical feature for accurate vehicle detection in congested scenes. YOLOv3 employs logistic loss and is built on the Darknet-53 network architecture [24], which provides a robust framework for processing visual data. Moreover, YOLOv3 utilizes nine anchor boxes to predict bounding boxes, an enhancement over the five anchors used in YOLOv2. This improvement significantly boosts the intersection over union (IoU) metric, leading to more precise and reliable object detection outcomes [25].

3.4. Object Tracking and Image Enhancement

Upon detecting a vehicle, a `TrackableObject` class is instantiated to represent a virtual entity linked to a specific target within a collection. This class is initialized with several parameters including the detected object's position at each estimation zone, the trajectory of its movement, and a boolean indicator signifying if the vehicle has crossed the final estimation point. Central to this object-tracking process is the `CentroidTracker`, a class dedicated to managing the lifecycle of each tracked object through functions like `register()`, `deregister()`, and `update()`. The `CentroidTracker` utilizes two ordered dictionaries to maintain a mapping of each object ID to its centroid location and to track how many consecutive frames the object has been absent, identified as “disappeared.”

To enhance image quality and reduce noise introduced during data capture—a common issue given the variable weather conditions—several preprocessing steps are implemented. Initially, images are resized and scaled to a uniform width to standardize the input data. Contrast-limited adaptive histogram equalization (CLAHE) is applied to improve the local contrast within the images while avoiding over-amplification. This method operates on discrete, small regions of the image, preventing the formation of artificial boundaries when adjacent tiles are merged [26].

Further, a Gaussian filter, known for its efficacy in smoothing, is employed to diminish noise within the images. Given the diversity of anomalies detected, including some identified from zoomed-in images, data augmentation strategies are essential to

enhance the robustness of object detection. Techniques such as photometric distortions (adjustments in brightness, hue, and saturation), geometric distortions (alterations in image dimensions), image expansion, and random cropping are utilized to substantially enrich the dataset diversity [27, 28]. These augmentation techniques not only create a more varied set of training data but also simulate different environmental conditions that might not have been directly captured in the initial dataset. The augmented datasets are detailed in Table 3, providing a comprehensive overview of the images available for each class of anomalies.

Table 3. Number of images for the augmented dataset.

Anomaly	Number of images/data
Standing on yellow box	3500
Traffic light signs	3250
Distance between vehicles	3000
Speeding	3550

3.5. Methodology

A CNN is a specialized type of neural network that incorporates one or more convolutional layers among other layers. The architecture of a CNN typically includes three key types of layers: the convolution layer, the pooling layer, and the fully connected layer. The convolution layer, fundamental to the network's operation, employs various filters to convolve the images derived from video feeds, extracting meaningful features while maintaining spatial relationships between pixels. The pooling layer, utilized for reducing the spatial dimensions (width, height) of the input volume for the next convolution layer, helps in achieving translational invariance and reducing the computational complexity by downsizing the feature maps. The fully connected layer follows the pooling layer and is responsible for outputting an N-dimensional vector, which serves as the basis for classification [29].

Transfer learning, which leverages a pre-existing model trained on a large dataset for a different but related task, is utilized in this project to enhance the efficiency of anomaly detection systems without the need to develop a model from scratch. This technique is particularly advantageous when annotated data are scarce or when computational resources are limited.

In this study, we have selected the MobileNet, VGG19, and ResNet50 models due to their proven effectiveness across various domains and a favorable balance between

classification accuracy and computational demand [30–32]. MobileNet, in particular, is designed as a “lightweight” variant of the conventional CNN architectures, making it ideal for deployment on devices with limited computational power. Despite its reduced parameter count, which generally leads to faster processing speeds at the expense of some accuracy, MobileNet remains effective for real-time applications [33]. It employs depthwise separable convolutions to decrease the number of parameters compared to a standard CNN of equivalent depth. This design includes a single 3×3 convolution layer that is followed by batch normalization and rectified linear unit (ReLU) activation, diverging from the traditional approach of combining 3×3 depth-wise convolution with 1×1 pointwise convolution. Figure 2 illustrates the deployment of depth-wise separable convolution in the MobileNet architecture.

Figure 2

[Open in figure viewer](#)[PowerPoint](#)

MobileNet architecture.

In our study, we employed the VGG19 architecture, a variant of the VGG model that includes 19 convolutional layers, 5 MaxPool layers, and 1 SoftMax layer. This architecture was initially proposed by Simonyan and Zisserman [34] and is renowned for its depth and robustness in image processing. The VGG19 was trained using the extensive ImageNet ILSVRC dataset, which comprises 1.3 million images distributed across 1000 classes. From this dataset, 100,000 images were designated for training and 50,000 for validation purposes [35].

The design of VGG19 utilizes small convolutional filters, specifically 3×3 , which is the smallest size to capture the notion of left/right, up/down, and center. This small filter size allows for deeper networks without a significant increase in computational complexity. In addition to the typical layers, VGG19 employs 1×1 convolution filters that serve as linear transformations of the inputs, with each followed by a ReLU to introduce nonlinearity. The convolutional stride is consistently set at one pixel, ensuring that the spatial resolution is maintained postconvolution. Notably, every hidden layer within the VGG network applies a ReLU activation function. The architecture concludes with three fully connected layers, further enhancing its capability to classify complex image data [36]. Figure 3 illustrates the detailed layer configuration of the VGG19 model.

Figure 3

[Open in figure viewer](#)[PowerPoint](#)

Architecture of VGG19.

ResNet50, yet another pretrained CNN model, was used to develop the automated anomaly detection application. ResNet consisted of 48 convolution layers, 1 MaxPool layer, and 1 AveragePool layer and was thoroughly trained with the ImageNet dataset. One benefit gained from the ResNet50 is that the model has an additional weight matrix, and thus, the skip connection enables direct signal propagation from the first to the last layer [37]. When the gradients are propagated directly from the loss layer to any other previous layer, the gradient signals can deteriorate. When the vehicle images are loaded, it goes through a convolution layer with 64 filters and a kernel size of 7×7 . The experiments carried out found that ResNet 50 generalizes better than the pretrained models and is tolerant to hyperparameters [38]. The residual module can identify the vehicles' high, medium and low levels of features and other features like traffic lights and yellow boxes. Figure 4 depicts the design of the ResNet50.

Figure 4

[Open in figure viewer](#)[PowerPoint](#)

Design of ResNet50.

The architecture of pretrained models provided a foundational framework for the development of our advanced CNN models. Traditionally, images are represented with dimensions of 224×224 . However, to better capture spatial and color information, we expanded the image representation to a three-channel format: $224 \times 224 \times 3$, incorporating width, height, and channel depth.

In the MobileNet architecture, we opted to utilize separable convolutions instead of the conventional depthwise separable convolutions. This adaptation integrated both depthwise and pointwise convolutions into a single layer, structured as follows:

(1)

This modification reduced the overall number of layers in the network to 14, thereby streamlining the architecture while retaining the functional advantages of depthwise separable convolutions. Despite these structural efficiencies, these changes did not significantly enhance the model's accuracy in detecting road anomalies.

Additionally, we investigated the effects of various regularization techniques on model performance, as regularizations can play a pivotal role in improving generalization. These techniques are known to be more effective when combined synergistically rather

than used in isolation [39]. To explore the impact on overfitting and performance, we substituted the standard ReLU activation function with dropout activation in certain layers:

(2)

This approach aimed to mitigate potential overfitting by creating variations in the activation during training, hence allowing us to assess discrepancies between training and testing accuracies across various anomaly classes.

Regarding the ResNet50 architecture, modifications were introduced to better suit vehicle anomaly detection tasks. The ResNet model typically employs two types of shortcut connections:

- 1.

Identity shortcut, where the input dimension equals the output dimension, thus bypassing the convolution layer.

- 2.

Convolutional shortcut, where a convolution is applied to match an increased output dimension.

In our custom adaptation, we selectively removed certain shortcut connections and incorporated additional classification layers tailored specifically to our defined anomaly classes. This strategic adjustment was aimed at optimizing the network by increasing the relevance and number of learnable parameters:

(3)

Table 4 provides a detailed overview of the parameter counts in the modified ResNet subnetworks, illustrating the structural adjustments and their implications for the model's capacity to learn and generalize across different anomaly detection scenarios.

Table 4. Number of parameters in the ResNet50 subnetworks.

ResNet50 subnetworks	Total parameters	Number of trainable parameters
Layer 37: last two identity blocks, the last convolution block are removed	78,96,123	76,98,523
Layer 40: last two identity blocks and the last convolution block are removed	95,96,952	91,23,865

ResNet50 subnetworks	Total parameters	Number of trainable parameters
Layer 43: last two identity blocks are removed	1,47,35,621	1,41,02,300
Layer 46: last identity block is removed	1,92,03,110	1,82,53,416
Layer 49: no module is removed	2,75,95,918	2,25,53,788

In this research, the adaptive moment estimation (ADAM) optimization algorithm is employed to enhance model training. ADAM calculates adaptive learning rates for each parameter, a feature particularly advantageous in complex networks where different segments exhibit varying sensitivities to weight adjustments. This adaptability is crucial for training deep neural networks effectively, especially in anomaly detection where different anomalies impact distinct regions of the input space [40].

The mathematical formulation of ADAM is given as follows:

Let g_t denote the gradient of the objective function with respect to the parameter θ at timestep t . ADAM maintains two variables, m_t and v_t , which are estimates of the first moment (the mean) and the second moment (the uncentered variance) of the gradients, respectively. The updates for these variables are as follows:

(4)

(5)

where β_1 and β_2 are hyperparameters that control the exponential decay rates of these moving averages. The initial values of m_t and v_t are typically set close to 0. To correct the initial bias toward zero, ADAM uses bias-corrected versions of these estimates:

(6)

(7)

Using these bias-corrected estimates, ADAM updates the parameters θ as follows:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t, \quad (8)$$

where η represents the step size (learning rate), and ϵ is a small constant added to ensure numerical stability.

This adaptive learning rate mechanism allows the model to adjust more effectively to the specific demands of different regions of the dataset, enhancing the ability to detect and classify complex behavioral patterns. The original pretrained and enhanced models were applied to our datasets, and the results were analyzed.

4. Results and Discussions

Anomaly detection involving speeding and stopping in a no-parking zone requires vehicle detection and tracking in a specific region of interest (ROI), referred to as the estimation zone. Figure 5a,b displays one example of the estimation zone of the region before starting any process.

Figure 5 (a)

[Open in figure viewerPowerPoint](#)

(a) Estimation lines on a highway and (b) estimation lines on a no parking zone.

Figure 5 (b)

[Open in figure viewerPowerPoint](#)

(a) Estimation lines on a highway and (b) estimation lines on a no parking zone.

In our study, the estimation zone for a specific region was defined by a set of y-coordinates, accompanied by data such as model paths, the real-world distance of the estimation zone, and the designated number of frames for performing object tracking as opposed to mere object detection. Once an object was detected with a confidence level of at least 30%, a CentroidTracker was initialized, and the object was cataloged in a list of TrackableObjects. Depending on the track_object variable value specified in the JSON configuration file, the object was tracked over several frames.

During the streaming of frames, the list of trackable objects was continually processed to analyze the position of each detected object in every frame. If a vehicle was assigned a tracker and it passed a predetermined estimation point, its position and the corresponding timestamp were recorded at that location, along with its unique object ID, in a CSV file.

Calculating the precise distance between two vehicles involved complex data extraction, primarily due to the difficulty of standardizing the conversion of pixel distances into real-life measurements. This metric varies across videos, necessitating a one-time calibration for each new location evaluated by the system. An approximation of the distance was achieved by estimating the distance between two known points

within a frame and then converting this measurement into pixel distance. For example, if it was established that the distance between two poles within a frame spanned 2 m and these poles occupied 50% of the screen width in a video with a resolution of 1920 ×

1080p, the pixel distance between the poles would be calculated as pixels. Given that 540 pixels correspond to 2 m, the distance between each vehicle could then be approximated.

All collected data were input into our object detection model, YOLOv3. This model underwent individual analysis to determine its capability in detecting objects across distances using the IoU metric. The results of these analyses are summarized in Table 5, which provides a detailed view of the detection capabilities and the effectiveness of the distance estimation methodology implemented in this study.

Table 5. Object detection results.

Types of objects	IoU (intersection of union) (%)
Vehicles	99.9
Traffic lights	98.6
Yellow box	99.8

The YOLOv3 model demonstrated outstanding performance in object detection, successfully identifying targets across diverse environmental conditions. It is important to highlight that the videos were captured under varying weather conditions, leading to fluctuations in the quality of the extracted images.

We incorporated both original pretrained models and their enhanced versions in our experiments. The dataset was partitioned into 70% for training, 10% for validation, and 20% for testing, enabling a balanced approach to model training and evaluation. To assess the performance of these models, we utilized classification accuracy as our primary metric. Classification accuracy is defined as the ratio of correct predictions to the total number of predictions made, providing a straightforward measure of model efficacy [41].

Figure 6 illustrates the training and validation accuracies achieved using the pretrained models on our dataset. This visual representation offers insight into the models' ability to generalize from the training data to unseen validation data, reflecting their robustness under varied imaging conditions. The graph also allows for the evaluation of overfitting or underfitting by comparing training accuracy against validation accuracy.

The discussion of these results highlights the effectiveness of YOLOv3 under the tested conditions and underscores the importance of using enhanced model variants to possibly improve detection accuracy in challenging visual environments. Further investigation into model parameters and training strategies could provide deeper insights into optimizing model performance across all segments of the dataset.

Figure 6

[Open in figure viewer](#)[PowerPoint](#)

Testing accuracy and training accuracy on the original dataset.

It was observed that the models were overfitting; that is, the models could not generalize well on our datasets. Therefore, before fine-tuning the models, we applied them to our augmented datasets. Figure 7 illustrates the accuracies of the augmented datasets.

Figure 7

[Open in figure viewer](#)[PowerPoint](#)

Testing accuracy and training accuracy on the augmented dataset.

From the experiments, it was noticed that the training accuracy improved. This shows that data augmentation does have some improvement on deep learning models. Furthermore, it was reported by Victor Ikechukwu et al. [32] that data augmentation deals with the class imbalance and overcomes the problem of poor generalization. This can also be confirmed in our case that data augmentation does improve the accuracies of the deep learning model. In addition, since this domain does not have a public dataset available for research, the number of annotated datasets is limited. It can also be observed that ResNet50 performed better compared to MobileNet and VGG19. This is the case since ResNet models generalize well based on their architecture [37].

We have applied the enhanced models, the MobileNet, ResNet50, and VGG19, on our augmented datasets. Figure 8 depicts the results.

Figure 8

[Open in figure viewer](#)[PowerPoint](#)

Testing accuracy on original architecture and enhanced architecture.

Initially, MobileNet was trained from scratch with 80 epochs without using any pretrained weights. We have varied the width parameter to experiment with the impact of latency versus accuracy. For our modified MobileNet, we have applied the separable convolutions instead of depthwise separable convolutions. Thus, the Keras framework combines both the depth and pointwise layers into one layer. This enhanced architecture reduced the network layers. However, the outputs concluded that it does not increase the accuracy significantly. Results show that the accuracy increased by only 1%. However, it is to be noted that the model takes, on average, 2 h less for the training than the ResNet50 and VGG19. During the experiments, it was also observed that the drop-activation technique increases the model's accuracy to 84% and reduces the overfitting problem present in the baseline architecture. Our enhanced ResNet model has achieved a testing accuracy of 95% compared to 90%. It is, thus, confirmed that the elimination of the shortcut modules and the addition of the classification layers had an impact on the overall performance, as also confirmed in [42]. This process has obtained the most relevant parameters, showing a positive impact.

Finally, the improved VGG19 was applied to our augmented dataset, achieving a testing accuracy of 90%. The adaptive learning rate, which continuously adjusted the weight based on the situation, had a positive effect on the system's overall performance. Thus, the concept of considering drop-activation reduces overfitting and improves generalization. As a result, our enhanced ResNet50 achieved a performance of 95%. To further analyze the performance of automated anomaly detection, it is also good to look at the individual performance of each anomaly. Table 6 summarizes the results achieved.

Table 6. Accuracy for individual classes.

Type of anomaly	Testing accuracy on the augmented datasets (%)
Overspeeding	95.3
Excessive stay on the yellow box	98.6
Improper distance between two vehicles	90.4
Nonconformance of traffic lights	98.3

The YOLOv3 model demonstrated outstanding performance in object detection, successfully identifying targets across diverse environmental conditions. It is important to highlight that the videos were captured under varying weather conditions, leading to fluctuations in the quality of the extracted images.

We incorporated both original pretrained models and their enhanced versions in our experiments. The dataset was partitioned into 70% for training, 10% for validation, and 20% for testing, enabling a balanced approach to model training and evaluation. To assess the performance of these models, we utilized classification accuracy as our primary metric. Classification accuracy is defined as the ratio of correct predictions to the total number of predictions made, providing a straightforward measure of model efficacy [41].

Figure 6 illustrates the training and validation accuracies achieved using the pretrained models on our dataset. This visual representation offers insight into the models' ability to generalize from the training data to unseen validation data, reflecting their robustness under varied imaging conditions. The graph also allows for the evaluation of overfitting or underfitting by comparing training accuracy against validation accuracy.

The discussion of these results highlights the effectiveness of YOLOv3 under the tested conditions and underscores the importance of using enhanced model variants to possibly improve detection accuracy in challenging visual environments. Further investigation into model parameters and training strategies could provide deeper insights into optimizing model performance across all segments of the dataset.

4.1. Ablation Study and Comparison With Other Models

To further understand the contributions of our model's design, we conducted an ablation study, which systematically evaluates the effects of key architectural changes on the model's performance. Specifically, we focused on eliminating shortcut modules and adding classification layers to the ResNet50 model. Our goal was to assess the impact of these changes on anomaly detection tasks like overspeeding, improper vehicle distance, and traffic light violations. Effects of removing shortcut modules.

Removing shortcut modules in our enhanced ResNet50 architecture allowed the model to concentrate on learning deeper representations of complex traffic anomalies. This change improved the model's ability to generalize across different anomaly classes, leading to a performance increase in overall accuracy from 90% to 95%. This improvement demonstrates the model's enhanced ability to learn features specific to road traffic anomalies.

4.1.1. Effects of Adding Classification Layers

Our model could perform more detailed and accurate predictions by integrating additional classification layers. This adjustment proved particularly beneficial for tasks requiring multiclass classification, such as distinguishing between traffic anomalies. The additional layers enabled more refined predictions, further contributing to the model's overall success in achieving high classification accuracy.

4.1.2. Comparison With Baseline Models

We compared the results of our enhanced ResNet50 model with two other baseline models, MobileNet and VGG19, which are commonly used in anomaly detection tasks. As shown in Table 7, our enhanced ResNet50 significantly outperformed both MobileNet and VGG19 across all key evaluation metrics. Notably, while MobileNet achieved 84% accuracy and VGG19 reached 90%, our enhanced ResNet50 achieved an impressive 95%. This comparison clearly demonstrates the superior performance of our enhanced architecture in detecting and classifying road traffic anomalies.

Table 7. Comparison of proposed model with existing literature.

Authors	Methods and performance	Discussion/limitations
Patil and Biswas [14]	Data mining approaches such as gradient boosting and decision tree	Have classified anomalies into no anomaly, unlevelled manhole, short bump and long bump. Have achieved a classification accuracy of 92.8%. Works well on labeled dataset. However, labeled datasets are not usually available for the types of anomalies like long stand on yellow boxes and speeding. Accuracy heavily depends on the labeled datasets
Ali, Atia, and Mostafa [18]	<i>k</i> -Nearest neighbor and dynamic time warping algorithms achieving an accuracy of 96.75%	Classification of drivers into normal and abnormal only. No consideration on the types of anomalies
Datondji et al. [43]	Hidden Markov model (HMM) achieving a performance of 93%	Traffic monitoring at intersection only, where bumping, passing, and jamming were being detected. However, the work has worked on only one anomaly

Authors	Methods and performance	Discussion/limitations
Maha Vishnu et al. [44]	Hybrid support vector machine (SVM with extended Kalman filter)	SVM was used to chase the vehicle. However, the abnormal behavior was to track the movement of the vehicle with respect to the traffic control
Our proposed model	Enhanced deep neural network for an overall performance of 95%	Multiclass classification of abnormal behaviors of drivers. Tracking and monitoring of vehicles in real life with respect to road compliance. Have used both labeled and unlabeled data. Have developed an enhanced deep learning model and make the model more efficient by analyzing the number of layers, learning rate, and data augmentation

In Figure 9, for the anomaly where there is an excessive stay in the red box, our system has obtained an accuracy of 98.6%. As explained in earlier sections, this type of anomaly is detected by extracting every detected vehicle's time spent in the ROI, the red box, and classifying this timestamp. The extraction process is done in two phases. The first extraction is done between the first two fixed lines inside the ROI. If this timestamp is classified as anomalous, the system marks the vehicle's trajectory as an anomaly. Figure 10 shows one of the tests conducted in real-time with respect to images captured with a long stay on yellow boxes. It may also happen that a vehicle abruptly stops after the first two fixed lines but is still inside the ROI. Then, in this case, the system extracts a timestamp of 0 and marks the vehicle as an anomaly, leading to a false positive.

Figure 9

[Open in figure viewerPowerPoint](#)

Speed anomaly prediction results.

Figure 10

[Open in figure viewerPowerPoint](#)

Vehicle standing on the yellow cross lines.

Our system achieved an accuracy of 90.4% for detecting the improper distance between two vehicles. Figure 11 demonstrates a real-time detection of such an anomaly taken in a capital city, where a line has been drawn between the vehicles that are too close. It is to be noted our system could not detect around 10% of such anomalies. This is primarily the case when there are many such anomalies in a single frame.

Figure 11

[Open in figure viewer](#)[PowerPoint](#)

Vehicle distances calculated only between close vehicles.

An accuracy of 98.3% was reached for the nonconformance of traffic lights anomaly. Finally, we have compared our work with some existing applications, which is summarized in Table 7.

5. Conclusion

Anomalous or uncommon behavior detection in road traffic is both a critical and sensitive topic for its real-world implementation. Implementing such a system reduces the risks of road accidents and discourages dangerous and reckless driving. To develop an automated vehicle anomaly detection application, we first captured videos in different weather conditions from which we have extracted frames to create our datasets. The anomalies and the vehicles were detected using the YOLOV3, which was able to capture even overlapping small objects. To overcome the issues of class imbalance and the issue of poor generalization of unseen data, we have applied photometric distortions, geometric distortions, expanded images, and random crops as data augmentation. In this work, we have selected the three pretrained models, the MobileNet, ResNet50, and VGG19 and performed a transfer learning on both of our original and augmented datasets. We have enhanced the three models by eliminating some shortcut modules and adding a set of classification layers based on our classes. The relevant learnable parameters are obtained during the training process. The ablation study and comparison with other algorithms clearly demonstrate that our enhancements to the ResNet50 architecture, specifically removing shortcut modules and adding classification layers, resulted in significant performance improvements. These enhancements allowed the model to more effectively detect and classify complex traffic anomalies, confirming the effectiveness of the modifications. In comparison to baseline models, the enhanced ResNet50 achieved superior accuracy, underscoring the robustness and applicability of our approach.

Additionally, we have computed the adaptive learning rates for each parameter. We have observed that our enhanced ResNet50 produced an overall accuracy of 95%. The individual classes were also analyzed, whereby an accuracy of 95.3% was achieved for overspeeding, 98.6% for an excessive stay on the yellow boxes, 90.4% with an improper distance between two vehicles, and 98.3% for the nonconformance of traffic lights. We have experienced several challenges like images with very complex scenes consisting of several anomalies considered in particular frames. There are scopes in the development of these kinds of applications. Currently, there are very few applications developed that consider all the types of anomalies that have been investigated in this research. However, there are further avenues where we can use explainable AI to investigate further the contributing factors leading to an anomaly.